

Faisal Lawan

Control Systems Researcher · Safety-Critical Control · Multi-Agent Robotics · Embedded Systems

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[\[gh\] GitHub](#)

[\[in\] LinkedIn](#)

[\[ax\] arXiv](#)

Education

The University of Manchester

Apr. 2026 – Present (Expected 2030)

PhD, Electrical & Electronic Engineering

- **Scholarship:** School of Engineering PGR Scholarship (Fully Funded).
- **Research Focus:** Control Barrier Functions (CBFs), multi-agent control, safety-critical autonomy, nonlinear robust/adaptive control.

The University of Manchester

Jul. 2021 – Jul. 2025

BEng (Hons), Mechatronics Engineering with Industrial Experience

First Class

- **Dissertation:** Safety-critical consensus-based formation control of distributed mobile robots.
- Top 0.5% of the 2025 graduating cohort; Outstanding Achievement Award.
- Best Final Year Project & Best in EEE Department.
- **Awards:** IET Future Talent Boost Scholarship (2024), Intel Fearless Innovation Award (2024).

Experience

Doctoral Researcher

Apr. 2026 – Present

The University of Manchester

Manchester, UK

- Architecting provably safe and stable Control Barrier Function (CBF) methods for multi-agent autonomous systems operating in highly dynamic, uncertain environments.
- Designing nonlinear control algorithms optimised for real-time edge computing on robotic platforms with severe computational and actuator constraints.

Robotics Research Assistant

Aug. 2025 – Apr. 2026

The University of Manchester – RAICo Programme

Manchester, UK

- Engineered a safety-critical adaptive impedance control framework for nuclear decommissioning, fusing IT2-Fuzzy Logic with Nonsmooth CBFs to guarantee real-time safety under severe parametric uncertainty.
- Developed a unified soft-constrained quadratic program (QP) solved online at 54 μ s per control cycle.
- Proved Uniform Ultimate Boundedness via composite Lyapunov analysis and deployed the framework on 7-DOF KINOVA and Franka Emika manipulators.

Cohort 2.s

Feb. 2026 – Apr. 2026

FR8

Espoo, Finland

- Selected from a global pool for an intensive deep-tech builder programme focused on accelerating transformative, highly technical projects.
- Collaborated alongside tier-1 VCs, deep-tech founders, and domain experts in a high-performance prototyping environment.

FPGA Hardware Engineer

Jul. 2023 – Aug. 2024

Altera (formerly Intel PSG)

Marlow, UK

- Architected and validated an FPGA hardware accelerator for Google's Snappy compression algorithm in SystemVerilog, boosting compression ratio by 10% and throughput by 30% versus software baselines.
- Engineered a NIOS V embedded test platform and custom driver software to stress-test the compression core.
- Benchmarked and redesigned search algorithms for duplicate MAC-address detection in VirtIO network devices, yielding a 100 $\times$ computational efficiency improvement.

Publications

Lawan, F., Zhan, W., Pan, W., & Carrasco, J. (2026). "Constrained Torque-Level Formation Control of Differential-Drive Wheeled Mobile Robots via Consensus-Based Feedback Linearised MPC." *Towards Autonomous Robotic Systems (TAROS 2026)*. Accepted.

Lawan, F., Han, X., Carrasco, J., Lennox, B., & Cheng, X. (2026). "Safety-Critical Adaptive Impedance Control via Nonsmooth Control Barrier Functions under State and Input Constraints." *Under review*. [arXiv:2605.28367](https://arxiv.org/abs/2605.28367)

Projects

- Tessellate: Distributed Formation Control Platform** | [\[GitHub\]](#) | *Python, MPC, SOCP* Oct. 2025
- Built a real-time platform for distributed formation control of nonholonomic robots using a consensus-based feedback-linearised MPC, solved as a Second-Order Cone Program at 60 Hz.
 - Implemented a shape-aware multi-layered A* planner and run-time-reconfigurable communication graph, with a PySide6 telemetry UI.
- Robot Poker Dealer and Player** | [\[GitHub\]](#) Sep. 2025 – Present
- Building an end-to-end autonomous poker-playing physical system using a modified SO-101 robot arm.
 - Fusing real-time computer vision (YOLOv8) with ROS 2 manipulation and game-theoretic decision making.
- GUH Quant Trading Challenge (1st Place)** | [\[GitHub\]](#) | *Python, Algorithmic Trading* Nov. 2025
- Won 1st place (60% of the prize pool) in the Greater Uni Hackathon algorithmic-trading challenge.
 - Built a quantitative market-making trading bot for the Limex market-making challenge.
- Quanser Robotics Hackathon (1st Place)** | [\[GitHub\]](#) Nov. 2025
- Engineered the fastest autonomous ring-sorting algorithm on Quanser's 4-DOF robot arm.
 - Derived geometric inverse kinematics from first principles to achieve precise sub-threshold cycle times without standard IK libraries (internet was down!).
- Autonomous Line-Following Robot** | [\[GitHub\]](#) Sep. 2022 – Jul. 2023
- Led a team of four to design a custom PCB and C++ STM32 control stack, achieving 100% line-following accuracy and winning the ESP Innovative Features Prize.
- Swarmhack (1st Place)** | *Python, ROS, Multi-Agent Swarms* Nov. 2022
- Designed and deployed a distributed algorithm achieving 100% autonomous aggregation of randomly moving robots in an enclosed physical space.

Skills

Robotics & Control: ROS/ROS2, Safety-Critical Control (CBFs, NCBFs), Linear/Nonlinear Systems, Robust/Adaptive/Optimal Control (PID, LQR, H-infinity, MPC), Stability Analysis.

Programming: Python, C/C++, TypeScript, MATLAB/Simulink, Node.js

Hardware Description Languages: SystemVerilog, VHDL, Verilog

Embedded Platforms: STM32, ESP32, ARM, Altera & Xilinx FPGAs, Raspberry Pi

Robotic Platforms: KINOVA Gen3, Franka Emika Panda, Boston Dynamics SPOT, Quanser, SO-101 Arm